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AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 302 — Past Questions & Materials

by Renance

Course: CSC 302

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — CSC 302

Level 300 • First Semester • PastQuestion

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AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF PHYSICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

2016/2017 FIRST SEMESTER DEGREE EXAMINATION

CSC 302: INTRODUCTION TO OPERATING SYSTEM

INSTRUCTION: Answer Any FOUR Questions **TIME: 2¼ Hrs**

1(a) State the two levels at which threads operate in modern operating system (2 marks)
 (b) What is a single-stream batch monitor. How does it differ from the multi-stream batch. (5 marks)

2(a) Mention and explain any three characteristics of operating system. (9 marks)
 (b) Enumerate any four desirable features of the general purpose operating system you know (4 marks)

3(a) Explain your understanding of operating system. ✓
 (b) State and explain in details the two major functions of operating system (10 marks) ✓
 (c) What makes macro instruction most interesting in the development of an operating system. (5 marks)

4(a) The detection of deadlock conditions is also a system overhead. Discuss (5 marks)
 (b) The simplest possibility for avoidance of deadlock is to introduce an extra layer of software for requesting resources in addition to a certain amount of accounting. Discuss. (10 marks)
 (c) Explain the possible problems with your approach in 3(b) (5 marks)

5(a) Explain your understanding of Deadlock (3 marks)
 (b) The school of thought has it that "waiting and synchronization is not all sweetness and roses" Discuss (5 marks)
 (c) Mention and explain the essential requirements for deadlock to occur (6 marks)
 (d) Explain any three methods for handling deadlock situation (6 marks)

6(a) What is semaphore (2 marks) ✓
 (b) Explain the meaning of the following terms:
 (i) kernel (ii) process (iii) process control block (iv) scheduling (v) co-operative scheduling (vi) pre-emptive scheduling (vii) starvation (2 marks each)
 (c) State the requirements for switching from one program to another (4 marks)

7(a) What is context switching? (2 marks)
 (b) Explain your understanding of virtual memory (4 marks)
 (c) Mention and explain the two methods used by virtual memory to free up RAM when needed (4 marks)
 (d) Suppose a problem arises in a situation where the system support several programmes resident in memory simultaneously. Then it is possible that the addresses coded into one programme will already be in use by another. Explain any three possible options to resolve such situation. (10 marks)

Characteristics of operating system are:
 1) Concurrency
 2) Non-determinacy
 3) Sharing
 4) Long-term storage of data and information.

AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF PHYSICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION FOR 2018/2019 ACADEMIC SESSION

CSC 302: OPERATING SYSTEM**Time: 2HRS****Instruction: Answer QUESTION 1 and any other THREE Questions**

- 1(a) Explain your understanding of operating system (4 Marks) ✓
 (b) State and explain the two major functions of operating system. (7 Marks) ✓
 (c) The operating system acts a host for applications that are run on the machine. Discuss. (4 Marks) ✓
 Pg 5 (d) Operating systems offer a number of services to application programs and users which are usually accessible through application programming interfaces (API). State any four of these services you know. (4 Marks)
 Pg 2 (e) List any four common contemporary operating system you know. (2 Marks)
 (f) Distinguish between Deadlock and starvation. (4 Marks) ✓
 Pg 2/3 (a) State and explain any three classes of operating system (7 Marks)
 (b) Describe the significant of virtual machine in computing. (4 Marks) ✓
 (c) With the aid of a diagram, describe the layered structure of an operating system. (4 Marks)
 Pg 25 (a) Describe context switching as applied in operating system. (2 Marks)
 Pg 14 (b) Explain your understanding of cooperating processes. (3 Marks)
 Pg 14 (c) State and explain any four reasons for providing an environment that allows process cooperation. (4 Marks)
 Pg 15 (d) State and explain any three issues for cooperating processes. (6 Marks)
 Pg 11, 13, 134 (a) What are some of the common events responsible for "process creation"? What constitutes "[process termination]"? Discuss any five conditions for "process termination" (7 Marks)
 Pg 14 (b) Mention and explain any four requirements that memory management is intended to satisfy. (8 Marks)
 Pg 15 (a) What is a semaphore? (3 Marks) ✓
 (b) What is the critical section problem. (3 Marks)
 (c) State with explanation, each requirement that a solution to the critical section problem must satisfy. (6 Marks)
 Pg 15 (d) State the mutual exclusion principle. (3 Marks)
 6(a) What is an Interrupt? (2 Marks)
 (b) Explain four classes of interrupt (4 Marks)
 (c) With the aid of a diagram describe a simple interrupt process. (5 Marks)
 (d) What effect does an interrupt have when it is sent to the CPU? (4 Marks)



Interface

Coffee

Renanced

AMBROSE ALI UNIVERSITY, EKPOMA
FACULTY OF PHYSICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
SEMESTER DEGREE EXAMINATION FOR 2019/2020 ACADEMIC SESSION

302: OPERATING SYSTEM

Instructions: Attempt Question **ONE** and any **THREE** other Questions. **Time:** 2hrs

- a) Vividly explain your understanding of Operating System. ✓ (4marks)
- b) ~~Pg~~ List and explain any three types of operating system you know. ✓ (6marks)
- (c) Describe the significant of a Virtual machine in computer science ✓ (3marks)
- (d) Considering the fact that operating system shields the user of the machine from the low-level details of the machine operation and provide frequently needed resources; explain the key elements responsible for this task. (6 marks)
- (e) Give a simple algorithm for achieving mutual exclusive between two processes with PID = 0 or 1. (6 marks)
- ~~Pg 2~~ 2(a) State and explain the two major functions of Operating System in details. ✓ (6 marks)
- (b) ~~Pg~~ Mention and explain the functions of the three pieces of programs available in the system Nucleus. (6 marks)
- ~~Pg 11~~ (c) Describe the implementation of a lock operation, using semaphores. (3 marks)
- 3(a) Describe deadlock as it relates to Operating System. (3 marks)
- (b) Indicate and explain the three methods required for handling deadlock situation. (6 marks)
- (c) Explain vividly, the role of Operating System shell and kernel to the computer. (6marks)
- ~~Pg 54~~ (a) What are System Calls? (3 marks)
- (b) ~~*~~ Mention and explain any three algorithms for short-term scheduling. (9marks)
- (c) Distinguish between Deadlock and Starvation. (3 marks)
- ~~Pg 295~~ (a) Describe any three of the essential requirements for a deadlock to occur (6marks)
- ~~3 29~~ (b) What strategies do deadlock avoidance and deadlock prevention algorithms deploy? (5 marks)
- ~~Pg 30~~ (c) Explain how the hold-and-wait condition can be prevented. (4 marks)
- ~~Pg 7~~ 6(a) Explain any four of the requirements that memory management is intended to satisfy in the computer. ✓ (6marks)
- (b) Explain your understanding of each of the following memory partition selection Algorithms. (i) First-fit (ii) Best-fit (ii) Worst-fit (6 marks)
- (c) State any three tasks of the memory management module of the operating system (3 marks)

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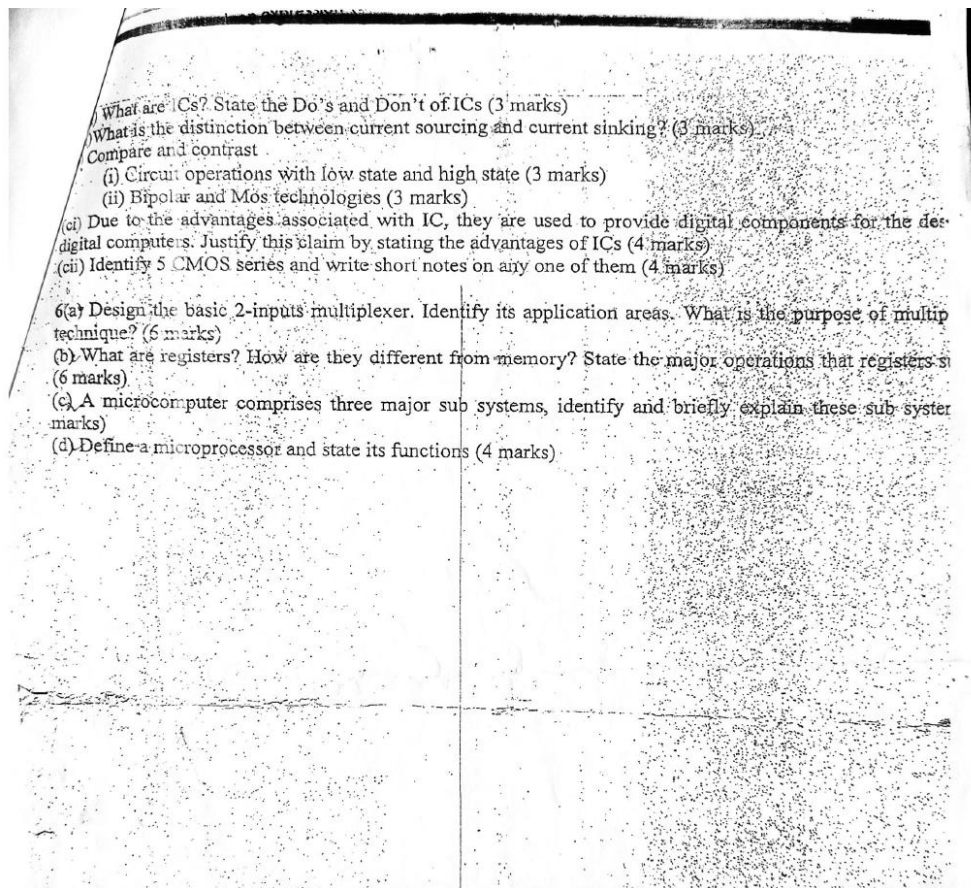
1ST SEMESTER DEGREE EXAMINATION FOR 2020/2021 ACADEMIC SESSION

CSC 302: INTRODUCTION TO OPERATING SYSTEM

Instruction: Attempt Question **ONE** and any **THREE** others

Time: 2hrs

- g2 1a. Explain your understanding of operating system. (3 marks)
 BS-b. State the importance of operating system utilities. (4 marks)
 Pa-c. Enumerate any five types of operating system you know (5 marks)
 Pa-d. Describe the functions of a distributed operating system. (3 marks)
 P3-c. State the advantages of Distributed operating system. (4 marks)
 P11 f. Differentiate between direct and indirect communication of a process. (4 marks)
 P5 & 16 g. How does a thread differ from a process? (2 marks)
- 2a. Write short note on each of the following:
 Loader (ii) Compiler (iii) Macro program (iv) Multi programming (v) Multi-tasking (vi)
 multi-user O/S (vii) Single-user O/S (14 marks)
- b. What is the slice time in a process? (1 marks)
- P9 23 3a. Switching between user program is done by part of the kernel. State the requirements for this to occur. (4 marks)
 P9 27 b. Design an algorithm for achieving mutual exclusions between two processes with PID = 0 or 1. (6 marks)
 P9 28 c. Explain what a system must do to recover from deadlock situation. (5 marks)
- 4a. Differentiate between stalemate and deadlock situation in process communication. (4 marks)
 10 & 29 29 b. (i) State and explain the essential requirements for a deadlock to occur. (3 marks)
 29 (ii) Indicate and explain the methods of resolution in deadlock situation (3 marks)
 * c. A process can guarantee itself uninterrupted access to shared data by switching off the scheduler. Discuss the drawbacks in this approach. (5 marks)
- 30 5a. Explain why the detection of deadlock conditions in process communication is described as system overhead. (5 marks)
 WA photo 6b. With the aid of a diagram only, describe the layered structure of an operating system. (2 marks)
- P5 c. Explain the significant of a virtual machine in computer science. (2 marks)
 P5 d. State and explain the two major functions of operating system. (6 marks)
- P6 6a. Define semaphore? (3 marks)
 P6 b. The concurrent process executing in the operating system may either be cooperating or independent processes. Discuss this in relation to process communication. (4 marks)
 P6 c. Indicate and explain any four reasons for providing an environment that allows process cooperation. (8 marks)



- What are ICs? State the Do's and Don't of ICs (3 marks)
- What is the distinction between current sourcing and current sinking? (3 marks)
- Compare and contrast
- (i) Circuit operations with low state and high state (3 marks)
 - (ii) Bipolar and MOS technologies (3 marks)
- (c) Due to the advantages associated with IC, they are used to provide digital components for the design of digital computers. Justify this claim by stating the advantages of ICs (4 marks)
- (d) Identify 5 CMOS series and write short notes on any one of them (4 marks)
- 6(a) Design the basic 2-inputs multiplexer. Identify its application areas. What is the purpose of multiplexing technique? (6 marks)
- (b) What are registers? How are they different from memory? State the major operations that registers support. (6 marks)
- (c) A microcomputer comprises three major sub systems, identify and briefly explain these sub systems (6 marks)
- (d) Define a microprocessor and state its functions (4 marks)

AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

FIRST SEMESTER DEGREE EXAMINATION FOR 2014/2015 ACADEMIC SESSION

CSC: 302: INTRODUCTION TO OPERATING SYSTEM

TIME: 2 Hours

Instruction: Answer QUESTIONS ONE and Any 3 others.

- 1 (a) Explain your understanding of operating system (4 marks)
(b) Explain the advantages and disadvantages a multi-user operating system (4 marks)
(c) Describe the two methods used by virtual memory to free-up RAM when required (6 marks) *Handwritten: paging & swapping. paging = dividing memory into small pages. swapping = moving pages from RAM to disk.*
- 2 (a) Describe the three methods for handling deadlock situation (6 marks) *Handwritten: Preventing, Detecting, Recovering.*
(b) what is Semaphore? (4 marks) *Handwritten: It is used to solve synchronization. It is a flag which can have a more general value than just 1.*
(c) Explain your understanding of the following: (i) Dispatching (ii) System overhead (iii) Context switching (iv) Quantum period (12 marks)
- 3 (a) What is process scheduling? (4 marks) *Handwritten: A regular time interval.*
(b) State the requirements for switching from one program to another (8 marks) *Handwritten: Interrupt events, saving & restoring programs and state.*
(c) Describe any two possible ways by which process scheduling can be realized (8 marks) *Handwritten: (1) Round-robin & priority scheduling. (2) First-come first-served.*
- 4 (a) Explain your understanding of a thread (4 marks)
(b) When is a task said to be multithreaded in an operating system (4 marks) *Handwritten: When a program is divided into multiple threads.*
(c) Describe the two levels at which threads operate in modern operating system (8 marks)
(d) Differentiate between direct and indirect communication of process (4 marks)
- 5 (a) Explain the three methods for handling deadlock situation (9 marks) *Handwritten: Preventing, Recovering, Detecting.*
(b) Explain the concept of shared libraries in program compilation (3 marks) *Handwritten: Shared libraries are the typical modules of an operating system.*
(c) What are the typical modules of an operating system (4 marks) *Handwritten: Kernel, Scheduler, File system, Process management.*
(d) Write short notes on any two of (c) above (4 marks)
- 6 (a) Explain the two major function of operating system (8 marks) *Handwritten: Technical layer of OS & Resource management.*
(b) Explain any three of the key elements available in operating system (6 marks) *Handwritten: File system, Command language, Device drivers.*
(c) Write a simple algorithm for achieving mutual exclusion between two processes with "I/N" equal to 0 or 1. (6 marks)

Handwritten notes:
1. A file system is a way of organizing files logically.
2. A simple command language enables user to write their own programs & manipulate files in a simple way.

Handwritten notes:
1. A state made of OS world. Deadlock occurs as a result of a wait-
ing for an event to occur which can be caused by an event.

Handwritten notes:
1. A set of processes wait for another process. Each waiting process
is a state of process. Resources can be shared to waiting process.
2. A semaphore is a variable that can be used to control access to a resource.
3. A semaphore can be used to control access to a resource.